MOHIT KARTHIKEYAN

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SUMMARY

Skilled in Java, SQL, Excel and data management. Experienced in the preparation of official correspondence and maintaining accurate financial records. Passionate about software development and problem-solving; project experience in deep learning, web development, and full-stack technologies. Seeking a position to utilize my technical and analytical skills in a software development or IT related role. Computer Operator at TNSTC Cooperative Thrift and Credit Society with practical knowledge of Computer Science and Engineering. Proficient in the use of dedicated society management software to maintain member records, financial accounts, demand registers and administrative operations.

SKILLS

Languages: Python, Java, JavaScript, SQL
Generative AI: LLM Development, RAG, Prompt Engineering, Vector Databases
AI/ML: Machine Learning, Data Analysis, Data Preprocessing, Scikit-learn, LightGBM, Pandas, NumPy, Predictive Analytics
Backend & APIs: FastAPI, REST APIs, Node.js, Express.js, Microservices
Frontend: React.js, Angular, HTML5, CSS3, Tailwind CSS
Tools: Git, GitHub, Google Colab, Jupyter Notebook, VS Code
Core: Data Structures & Algorithms, DBMS, Object-Oriented Programming

PROJECTS

Bone Fracture Detection based on Deep Learning Techniques *Deep Learning*

- Developed an end-to-end system for automatic bone fracture classification using advanced deep learning architectures like ResNet, DenseNet, VGG19, EfficientNet, and Vision Transformers.
- Applied preprocessing, data augmentation, and transfer learning techniques to enhance model performance and generalization. Designed and evaluated hybrid models, achieving highest accuracy with DenseNet-Hybrid.

Empowering Farmers with Real-Time Market Access *Flutter, Figma, Firebase*

- Created a digital platform connecting farmers directly with buyers using AI, blockchain, and data analytics for real-time market access.
- Implemented Aadhaar/OTP-based authentication and fraud detection to ensure secure transactions.
- Used tamper-proof blockchain records for transparent and trustworthy trade.
- Developed an admin panel for real-time monitoring, dispute resolution, and tracking market trends.
- Enhanced efficiency and trust in agri-commerce by fostering a farmer-centric ecosystem.

Enterprise Agentic RAG Assistant *Python, LangChain, LangGraph, RAG, LLMs, Vector Database, FastAPI*

- Developed an AI-powered enterprise document assistant using Python, LangChain, and LangGraph to generate grounded, context-aware responses.
- Implemented RAG with document ingestion, text chunking, embeddings, vector similarity search, and source attribution.
- Built an agentic tool-calling workflow and exposed it through FastAPI REST APIs with modular components, evaluation, logging, and basic guardrails.

INTERNSHIP

Training Intern – TapAcademy Certificate *Feb 2025 – Jul 2025*

- Attending training based on Java full stack.
- Gained hands-on experience in Java Full Stack Development. Learned core skills including Java, Data Structures and Algorithms (DSA), SQL, HTML/CSS, and JavaScript. Currently working with Hibernate and Spring Boot to understand backend frameworks and how to build complete full-stack projects from scratch.

WORKING EXPERIENCE

Computer Operator – TNSTC Cooperative Thrift and Credit Society *Jun 2026 – Present*

- Maintained and managed society records using specialized cooperative society software, including Chitta Book, Demand Register, Member Records, and Profit and Loss Accounts.
- Prepared official notices, letters, and communications regarding overdue payments and member-related matters.
- Managed and updated member databases and financial records using Microsoft Excel. Assisted in ensuring accurate record-keeping, reporting, and day-to-day administrative operations of the society.

EDUCATION

Institute of Cooperative Management *2025 – 2026*
Diploma in Cooperative Management *Erode*

Kongu Engineering College *2021 – 2025*
B.E. in Computer Science and Engineering, CGPA: 6.79 *Erode*

Kongu Vellalar Matriculation Higher Secondary School *2019 – 2021*
HSC, Percentage: 86.58% *Erode*

Kongu Vellalar Matriculation Higher Secondary School *2018 – 2019*
SSLC, Percentage: 81% *Erode*